

BOMA BRAIDE

bomab1@umbc.edu

(443) 869-0596

Baltimore, MD

github.com/bbraide

Motivated Computer Science graduate (UMBC '26, Cum Laude) combining systems programming and digital illustration with expertise in C++ and Python, demonstrated through ray tracing and game development projects, and an independent art following of 12.5k.

WORK EXPERIENCE

University of Maryland, Baltimore County -

Teaching Assistant — Introduction to Computer Science/Python

Jan 2024 - current

Baltimore, MD

- Led weekly lab sections and office hours for 30+ students, clarifying core CS concepts including data structures, recursion, and object-oriented design
- Collaborated with course instructors to align lab content with lecture material, ensuring a cohesive learning experience across all course sections
- Graded assignments and provided detailed written feedback, maintaining consistent academic standards across a large course section
- Mentored students one-on-one, fostering confidence and independent problem-solving skills in a rigorous introductory computer science curriculum

PROJECTS

C++ UNO Game - C++20

2024 - 2026

- **Polymorphic card hierarchy:** Designed and implemented a fully playable 2–4 player UNO game featuring a polymorphic card hierarchy — abstract base class Card with pure virtual CanPlayOn(), extended by NumberCard, ActionCard, and WildCard subclasses each resolving play validity independently
- **Circular doubly linked list:** Engineered a circular doubly linked list (Lineup) for turn order management, supporting real-time direction reversal, player skipping, and wrap-around with no nullptr checks required
- Built a custom ANSI RGB terminal UI (Colors.h) with per-card color theming, bold special cards, lavender current-player highlight, and a scoped enum class for type-safe color selection
- Implemented file I/O parsing (getline with delimiter), dynamic memory management with a safe destruction order, operator overloading, and AI thinking delay across 1,400+ lines of original code

C++ Ray Tracer - C++

Jan 2026 - Mar 2026

- Implemented a ray casting engine from scratch in C++ capable of rendering scenes containing hundreds of spheres and arbitrary polygon meshes parsed from Polyray (.pi) scene files
- Extended to full Phong illumination with ambient, diffuse, and specular shading; added recursive mirror reflection (up to 15 levels) using the Schlick–Fresnel approximation and gamma-corrected PPM output
- Implemented shadow testing via a secondary probe ray function and converted all vector mathematics and intersection algorithms (ray–sphere, ray–triangle) directly from equations to optimized C++ code

Unix-Style File System Simulator - Python

Nov 2022

- Built a fully interactive command-line file system simulator in Python, replicating core Unix shell commands: ls, cd, mkdir, touch, rm, locate, and pwd
- Modeled the directory tree as a recursive nested dictionary structure, supporting absolute paths, relative paths, and cd .. traversal entirely through recursive helper functions
- Implemented robust input validation, command history tracking, and descriptive error messages — earned a perfect score as a first-year CS student

Data Science End-to-End Analysis - Python

Spring 2026

- Applied full data science pipeline including data cleaning, exploratory analysis, feature engineering, and statistical modeling using pandas, NumPy, and matplotlib
- Produced visualizations and written analysis communicating findings from a real-world dataset to a non-technical audience

EDUCATION

University of Maryland, Baltimore County (UMBC) - Bachelor of Science, Computer Science, Cum Laude

August 2022 - May 2026 GPA: 3.6 / 4.0

Baltimore, MD

Awards

- **Outstanding Senior in Computer Science** (UMBC COEIT) | *May 2026* – Awarded for exceptional academic achievement and technical leadership within the department.

SKILLS

- Languages: C++, Python, Java
- Concepts: Object-Oriented Programming, Data Structures & Algorithms, Linked Lists, Recursion, File I/O, Memory Management, Ray Tracing, Computer Graphics, Systems Programming
- Tools: Visual Studio, Git/GitHub (version control), VS Code, pandas, NumPy, matplotlib
- Other: Technical instruction & mentorship (2 years TA), creative outreach — independent artist with 12.5k followers (@bberetd)